Practical Machine Learning: Session 1 Cheatsheet

1. Fundamentals of Information Theory

Self-Information

- **Definition**: $I(x) = -\log_2(p)$ where p is probability of event
- **Interpretation**: Measures "surprise" of an event
  - Rare events (small p) → high information
  - Common events (large p) → low information

Entropy

- **Definition**: $H(X) = -\sum_i p_i \log_2(p_i)$
- **Meaning**: Average uncertainty in a random variable
- **Properties**:
  - Minimum = 0 when one outcome has probability 1 (no uncertainty)
  - Maximum when all outcomes equally likely
  - Fair coin → $H = 1$ bit; biased coin → $H < 1$ bit

Joint and Conditional Entropy

- **Joint entropy** $H(X, Y)$: Uncertainty of two variables together
- **Conditional entropy** $H(Y|X) = H(X, Y) - H(X)$: Remaining uncertainty about Y when X is known
- **Key**: Knowing X reduces uncertainty about Y

Mutual Information

- **Definition**: $I(X; Y) = H(X) + H(Y) - H(X, Y)$
- **Interpretation**: How much information X gives about Y
- **Property**: $I(X; Y) = 0$ iff X and Y are independent
- **ML use**: Feature selection and information gain in decision trees

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2. Introduction to Machine Learning

Core Concept

- **Traditional programming**: Humans write rules → computer applies them
- **Machine learning**: Provide data + algorithm → algorithm learns rules from data

Typical Components

- **Input**: Data (features) x
- **Output**: Model f that maps $x \to \hat{y}$ (prediction)
- **Learning algorithm**: Searches hypothesis space to minimize loss/error
- **Hypothesis space**: Set of all models the algorithm can choose from

Types of ML Problems

1. **Supervised**: Input-output pairs (x, y) given
    - Classification: discrete labels
    - Regression: continuous values
2. **Unsupervised**: Only inputs x, no labels
    - Clustering, dimensionality reduction
3. **Reinforcement**: Agent learns from environment rewards/penalties
4. **Semi-supervised**: Few labeled + many unlabeled samples
5. **Self-supervised**: Creates own supervision signal from data

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3. Algorithm Types of Machine Learning

Supervised Learning

- **Data**: Input–output pairs (x, y) with known labels
- **Goal**: Learn mapping $x \to y$
- **Classification**: Discrete output (spam/not spam, disease/no disease)
- **Regression**: Continuous output (price, temperature, salary)
- **Analogy**: Learning with answer key from a teacher

Unsupervised Learning

- **Data**: Only inputs x, no labels
- **Goal**: Discover hidden structure
- **Clustering**: Group similar data (customer segmentation)
- **Dimensionality reduction**: Compress data (PCA)
- **Analogy**: Forming groups based only on observation

Reinforcement Learning

- **Mechanism**: Agent takes actions, receives rewards/penalties
- **Goal**: Learn policy to maximize cumulative reward
- **Components**: State, action, reward, policy
- **Examples**: Game-playing (chess, Go), robotics, adaptive recommendations
- **Analogy**: Learning to ride a cycle through trial-and-error

Online vs Batch Learning

- **Batch (offline)**: Train on entire fixed dataset once
- **Online**: Data arrives sequentially; model updates continuously
- **Use case**: Online for streaming data (stock prices, click streams)

Parametric vs Non-Parametric

- **Parametric**: Fixed number of parameters (linear regression, logistic regression)
- **Non-parametric**: Parameters grow with data (k-NN, decision trees)

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4. Probably Approximately Correct (PAC) Learning

Definition

A concept class is **PAC-learnable** if there exists an algorithm that, for any target concept and data distribution, outputs a hypothesis that is:

- **Approximately correct**: True error $\leq \epsilon$
- **Probably correct**: With probability $\geq 1 - \delta$

Key Parameters

- ϵ (epsilon): **Accuracy tolerance** – maximum allowed true error
  - Smaller ϵ → stricter accuracy → more samples needed
- δ (delta): **Failure probability** – allowed probability of exceeding error bound
  - $1 - \delta$ = confidence level (e.g., $\delta = 0.05$ → 95% confidence)

Sample Complexity

- Number of samples needed to achieve given ϵ, δ:

$$m \geq \frac{1}{\epsilon}\left(\log \frac{1}{\delta} + VC(H)\right)$$

- **Intuition**: Stricter requirements → more data needed

True Error vs Training Error

- **Training error**: Error on training set (empirical error)
- **True error**: Misclassification probability on new examples from true distribution
- **PAC focus**: Bound true error using finite training data

Distribution-Free PAC

- Learning guarantees hold for **any** underlying data distribution
- Does not assume specific distribution (e.g., Gaussian, uniform)

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5. Uses of Machine Learning

Domain Applications

Domain	Use Cases
Healthcare	Disease diagnosis, medical image analysis (X-ray, MRI), patient admission prediction
Finance	Fraud detection, credit scoring, loan approval, algorithmic trading
E-commerce	Product recommendations, customer segmentation, churn prediction
NLP	Machine translation, chatbots, sentiment analysis, question answering
Computer Vision	Image classification, object detection, face recognition, segmentation
Cybersecurity	Intrusion detection, malware classification, anomaly detection

Task Type Mapping

- **Classification**: Email spam detection, disease diagnosis, object recognition
- **Regression**: House price prediction, demand forecasting, temperature prediction
- **Clustering**: Customer segmentation, document grouping, anomaly detection
- **Dimensionality reduction**: Data visualization, feature compression

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6. Evaluating ML Techniques

Dataset Splits

- **Training set**: Fit model parameters
- **Validation set**: Tune hyperparameters and model selection
- **Test set**: Final performance estimation (used only once)
- **Analogy**: Practice exams (train) → mock exam (validation) → final board exam (test)

Overfitting vs Underfitting

- **Overfitting**: Model fits training data too well, poor on unseen data (memorization)
- **Underfitting**: Model too simple, poor even on training data

Confusion Matrix (Binary Classification)

	Actual Positive	Actual Negative
Predicted Pos.	TP	FP
Predicted Neg.	FN	TN

- **TP** (True Positive): Predicted positive, actually positive
- **TN** (True Negative): Predicted negative, actually negative
- **FP** (False Positive): Predicted positive, actually negative (Type I error)
- **FN** (False Negative): Predicted negative, actually positive (Type II error)

Core Metrics

Metric	Formula	Question It Answers
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	Of all predictions, how many are correct?
Precision	$\frac{TP}{TP+FP}$	Of all predicted positives, how many are actually positive?
Recall (Sensitivity)	$\frac{TP}{TP+FN}$	Of all actual positives, how many did we detect?
F1-score	$2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$	Harmonic mean of precision and recall
Specificity	$\frac{TN}{TN+FP}$	Of all actual negatives, how many were predicted negative?

When to Use Which Metric

Scenario	Best Metric
Balanced classes, equal costs	Accuracy
High cost of false positives (spam in inbox)	Precision
High cost of false negatives (missing fraud, cancer)	Recall or F1
Imbalanced data, need overall ranking	AUC-ROC

ROC and AUC

- **ROC curve**: Plots True Positive Rate (TPR) vs False Positive Rate (FPR) for various thresholds
- **AUC (Area Under Curve)**: Scalar 0–1; higher AUC = better overall ranking of positives vs negatives
- **Use**: Good for imbalanced data, evaluates classifier across all thresholds

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Quick MCQ Keywords

Concept	Key Terms
Information Theory	Entropy, mutual information, information gain, surprise
ML Types	Supervised (classification/regression), unsupervised (clustering), reinforcement
PAC Learning	ϵ (accuracy), δ (confidence), sample complexity, distribution-free
Evaluation	TP, FP, FN, TN, precision, recall, F1, AUC, confusion matrix, overfitting
Applications	Healthcare (diagnosis), finance (fraud), e-commerce (recommendation), NLP, vision

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Common MCQ Mistakes to Avoid

1. **Entropy ≠ 0 when outcomes are equally likely** → Entropy is MAXIMUM when equally likely
2. **PAC**: Confused which parameter is which → ϵ = accuracy, δ = confidence
3. **Precision vs Recall**: Mixed up the direction → Precision = predicted positive, Recall = actual positive
4. **False Positive vs False Negative**: FP = predicted yes but actually no; FN = predicted no but actually yes
5. **Accuracy on imbalanced data**: Don't use accuracy alone for imbalanced problems; use recall/F1/AUC
6. **Confusion matrix**: Remember rows/columns or TP location

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Practice Formula Summary

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$H(X) = -\sum_{i} p_i \log_2(p_i)$$

$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$

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